🎛️ Configuration

Model Info

Model Type: RandomForestRegressor

Analysis Options

- ☑ Show Feature Importance
- ☑ Show Sustainability Analysis
- ☑ Show Profitability Analysis
- ☑ Show Advanced Visualizations

📊 Chart Options

Chart Theme

plotly

☑ Show Debug Info

🌱 Sustainable Textile Demand Forecasting

Advanced ML-powered forecasting with comprehensive profitability analysis - **Universal Format Support**

✅ Model loaded successfully

📁 Data Upload - Universal Format Support

Upload your textile dataset

Drag and drop file here
Limit 200MB per file • CSV, XLSX, XLS, JSON, FARQUET

Browse files

enhanced_textile_data.csv 0.7MB

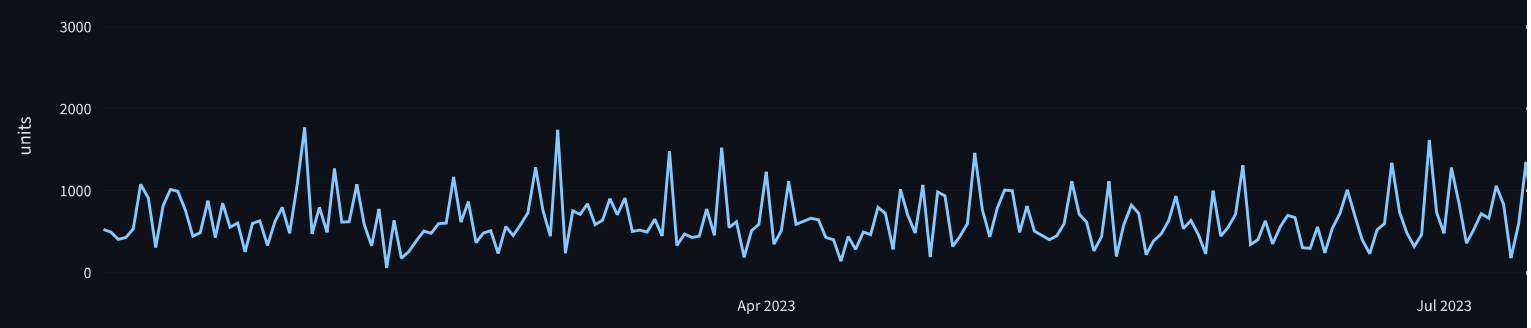
> 📄 File Information

✅ File processed successfully!

🔍 Quick Data Visualization Test

📊 Testing chart with column: **units**

TEST: units Over Time



✅ TEST CHART DISPLAYED SUCCESSFULLY!

📋 Dataset Analysis

Records	Columns	Date Range
8000	15	729 days

🔍 **Detected Columns:**

- ✅ date (detected)
- ✅ sku (detected)
- ✅ store (detected)
- ✅ units (detected)
- ✅ material (detected)
- ✅ category (detected)
- ✅ price (detected)
- ✅ revenue (detected)

📊 **Additional Columns Found:**

- initial_stock
- remaining_stock
- on_promotion
- customer_rating
- store_size
- sustainability_preference

📊 Data Preview

	date	sku	store	category	m
0	2024-02-13 00:00:00	TEX093	Chicago_Central	jeans	c
1	2024-09-11 00:00:00	TEX026	Miami_Beach	t-shirt	li
2	2023-11-15 00:00:00	TEX062	NYC_Flagship	jacket	p
3	2024-09-16 00:00:00	TEX070	NYC_Flagship	pants	h
4	2023-10-23 00:00:00	TEX027	NYC_Flagship	shorts	c
5	2023-04-08 00:00:00	TEX019	LA_Boutique	jacket	re
6	2024-08-14 00:00:00	TEX022	Denver_Green	dress	a
7	2024-08-13 00:00:00	TEX055	Boston_Harbor	jacket	re
8	2023-03-06 00:00:00	TEX050	NYC_Flagship	jeans	b
9	2024-08-18 00:00:00	TEX089	Boston_Harbor	shorts	li

✅ Adaptive features created successfully!

📊 Comprehensive Data Analysis & Visualizations

🔍 **Chart Debug Info:**

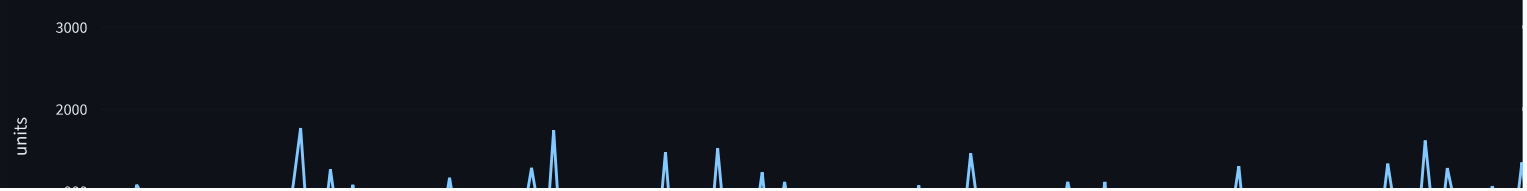
📊 DataFrame shape: (8000, 33)

📋 Available columns: ['date', 'sku', 'store', 'category', 'material', 'units', 'price', 'revenue', 'initial_stock', 'remaining_stock', 'on_promotion', 'customer_rating', 'is_sustainable', 'store_size', 'sustainability_preference', 'weekofyear', 'month', 'quarter', 'day_of_week', 'is_weekend', 'is_holiday_season', 'is_spring_summer', 'is_seasonal_category', 'units_lag_1', 'units_lag_2', 'units_lag_4', 'units_lag_8', 'units_ma_4', 'units_std_4', 'units_ma_8', 'units_std_8', 'units_ma_12', 'units_std_12']

📈 Data Overview

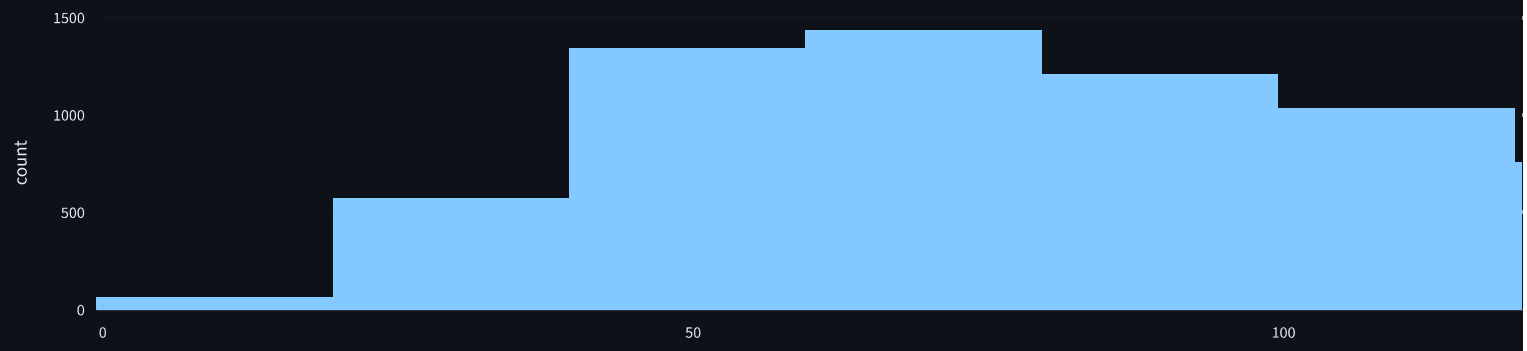
📊 Analyzing: **units**

Units Over Time



✅ Time series chart created!

Units Distribution

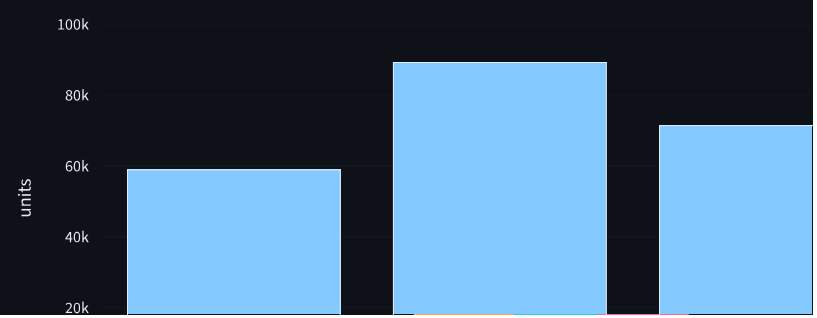


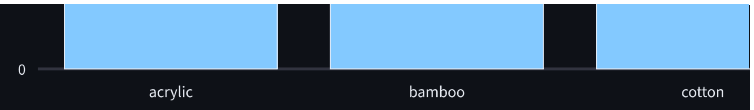
✅ Distribution chart created!

📝 Material Analysis

Units by Material

Units by Material Type





✅ Material pie chart created!

✅ Material bar chart created!

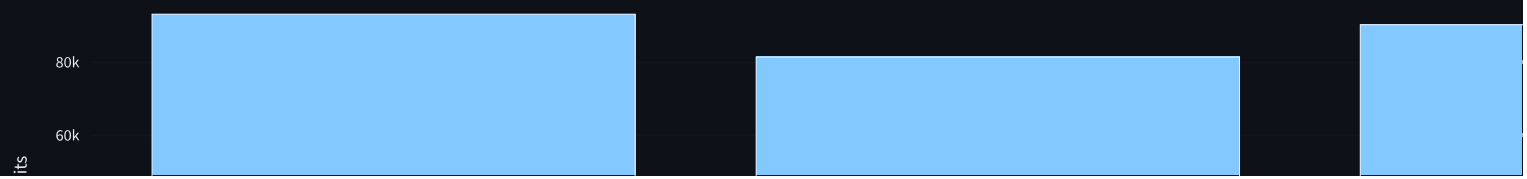
🏪 Store Analysis

Units by Store

✅ Store analysis chart created!

🏷️ Category Analysis

Units by Category



✅ Category analysis chart created!

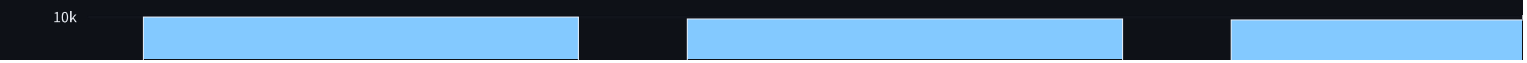
🔬 Data Relationships

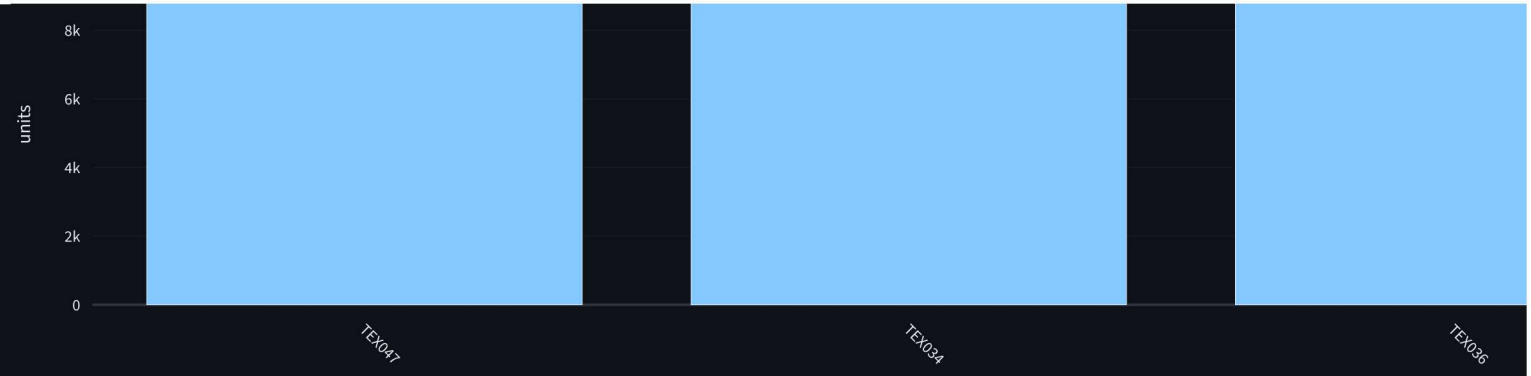
Feature Correlation Heatmap

✅ Correlation heatmap created!

🏆 Top Performing SKUs

Top 10 SKUs by Units





✅ Top SKUs chart created!

📋 Data Sample

	date	sku	store	category	material	units	price	revenue	initial_stock	rer
0	2023-01-02 00:00:00	TEX034	Denver_Green	jacket	hemp	100	135.53	13552.73	221	
1	2023-01-02 00:00:00	TEX062	NYC_Flagship	blouse	recycled_polyester	122	59.41	7248.48	254	
2	2023-01-02 00:00:00	TEX045	Seattle_Eco	skirt	linen	33	55.39	1827.71	53	
3	2023-01-02 00:00:00	TEX045	LA_Boutique	pants	organic_cotton	88	68.47	6025.63	158	
4	2023-01-02 00:00:00	TEX052	Chicago_Central	sweater	viscose	180	50.57	9103.44	486	
5	2023-01-03 00:00:00	TEX097	LA_Boutique	sweater	tencel	123	84.71	10419.16	153	
6	2023-01-03 00:00:00	TEX066	NYC_Flagship	skirt	organic_cotton	74	55.05	4073.91	197	
7	2023-01-03 00:00:00	TEX083	Boston_Harbor	dress	acrylic	53	69.32	3673.8	136	
8	2023-01-03 00:00:00	TEX080	Portland_Sustainable	jeans	tencel	84	104.69	8793.57	157	
9	2023-01-03 00:00:00	TEX004	Portland_Sustainable	pants	bamboo	63	88	5544.24	186	

✅ Data table displayed!

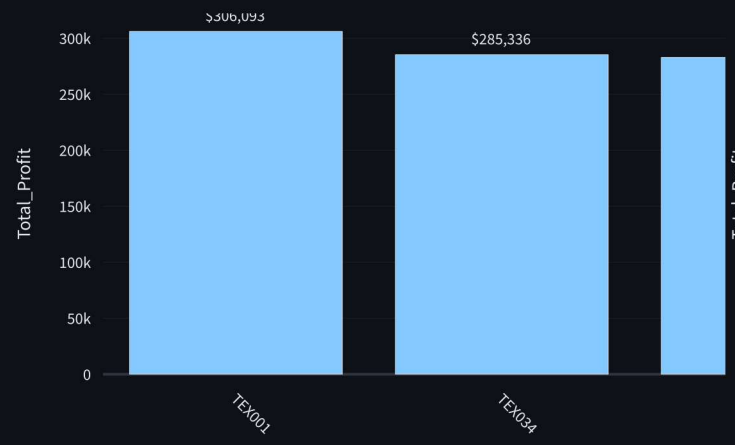
📈 Successfully generated 7 chart types: Time Series, Distribution, Material Analysis, Store Analysis, Category Analysis, Correlation, Top SKUs

💰 Profitability Analysis

Analysis Methods Used: Revenue & Price Analysis

🏆 Most Profitable Products

Top 10 Most Profitable SKUs



Bottom 10 Least Profitable SKUs

🥇 **Top Profitable SKUs:**

	SKU	Tot
0	TEX001	$3
33	TEX034	$2
28	TEX029	$2
68	TEX069	$2
35	TEX036	$2
40	TEX041	$2
53	TEX054	$2
65	TEX066	$2
46	TEX047	$2
8	TEX009	$2

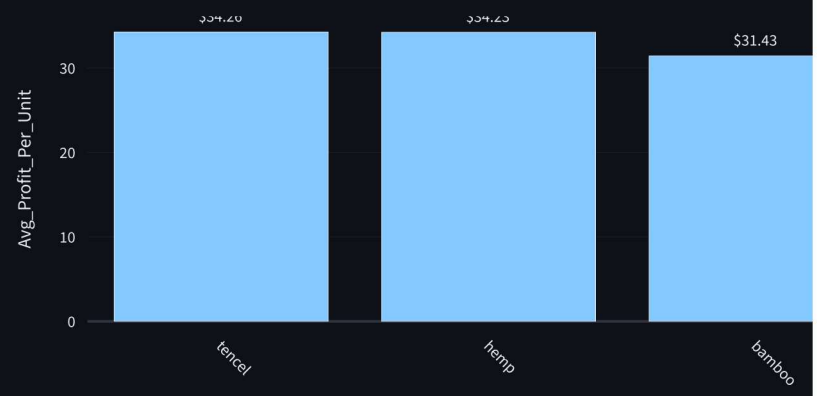
📉 **Least Profitable SKUs:**

	SKU	Total_Profit
70	TEX071	$179,589.58
85	TEX086	$177,029.98
45	TEX046	$176,127.18
99	TEX100	$172,780.44
17	TEX018	$171,929.48
55	TEX056	$171,767.73
39	TEX040	$165,737.18
90	TEX091	$165,480.67
1	TEX002	$159,408.81
3	TEX004	$157,169.17

📝 Profitability by Material Type

Profit Distribution by Material

Average Profit Per Unit by Material

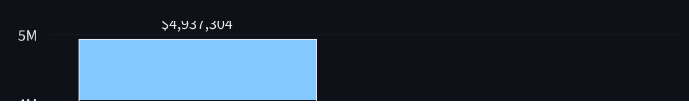


📊 **Material Profitability Summary:**

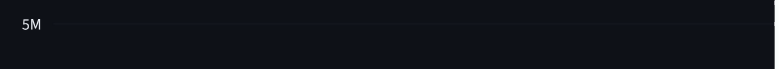
	Material	Total_Pr
8	tencel	$3,290,2
3	hemp	$2,958,8
1	bamboo	$2,762,2
5	organic_cotton	$2,682,3
7	recycled_polyester	$2,628,1
4	linen	$2,396,8
2	cotton	$1,710,9
9	viscose	$1,285,4
6	polyester	$1,243,5
0	acrylic	$1,066,8

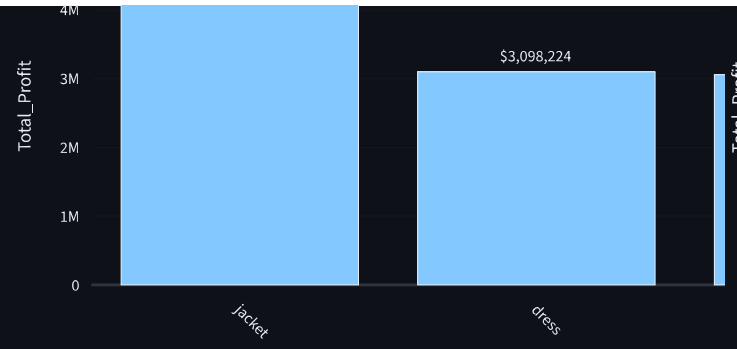
🏷️ Profitability by Product Category

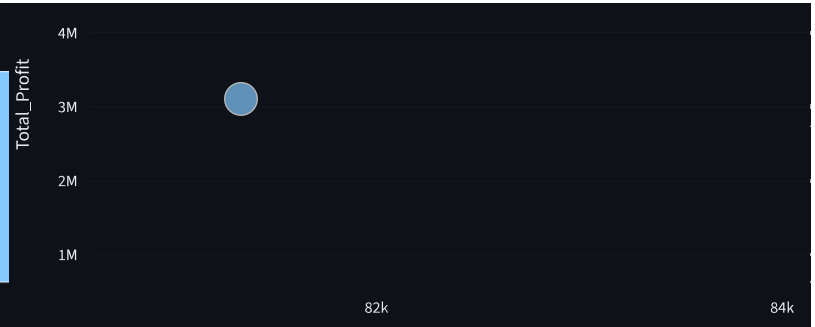
Total Profit by Product Category



Profit vs Volume by Category

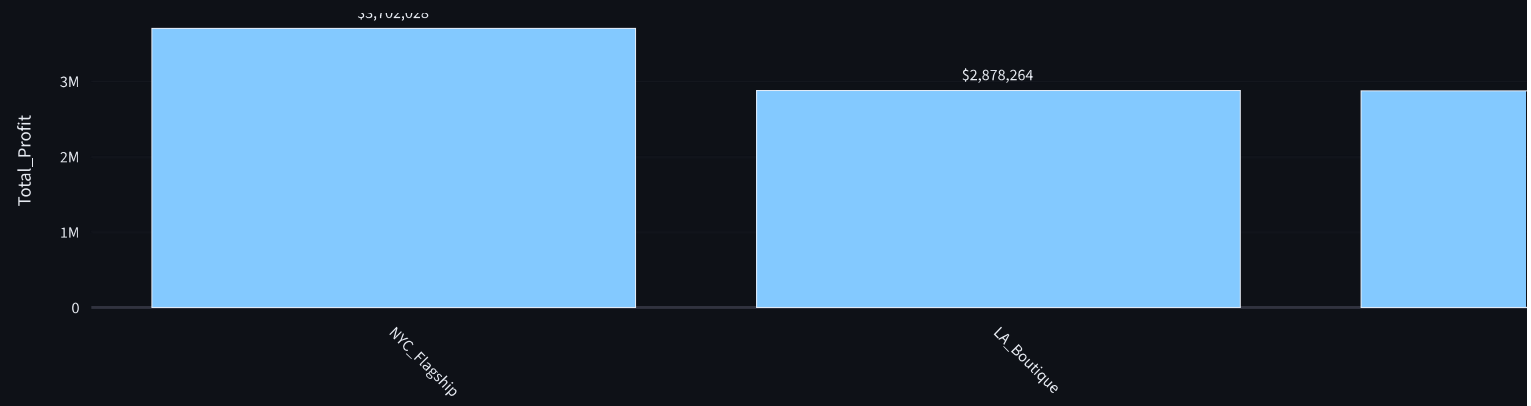






🏪 Store Profitability Analysis

Total Profit by Store

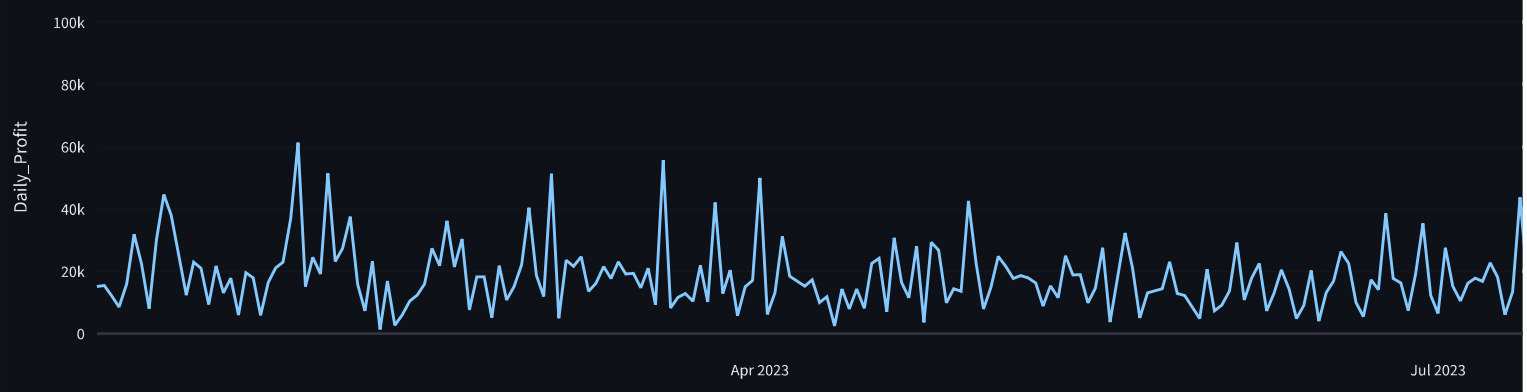


🏆 **Store Performance Ranking:**

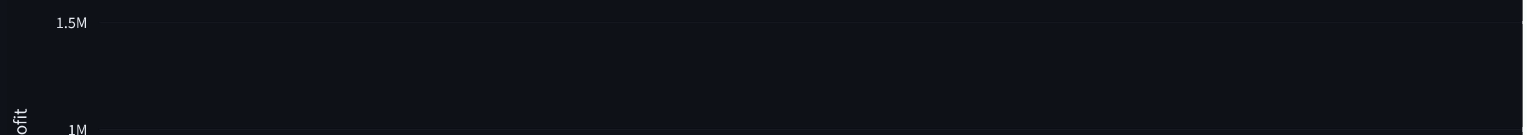
	Store	Total_
6	NYC_Flagship	$3,702
4	LA_Boutique	$2,878
2	Chicago_Central	$2,872
1	Boston_Harbor	$2,460
5	Miami_Beach	$2,178
0	Austin_Urban	$2,125
8	Seattle_Eco	$2,026
7	Portland_Sustainable	$1,977
3	Denver_Green	$1,804

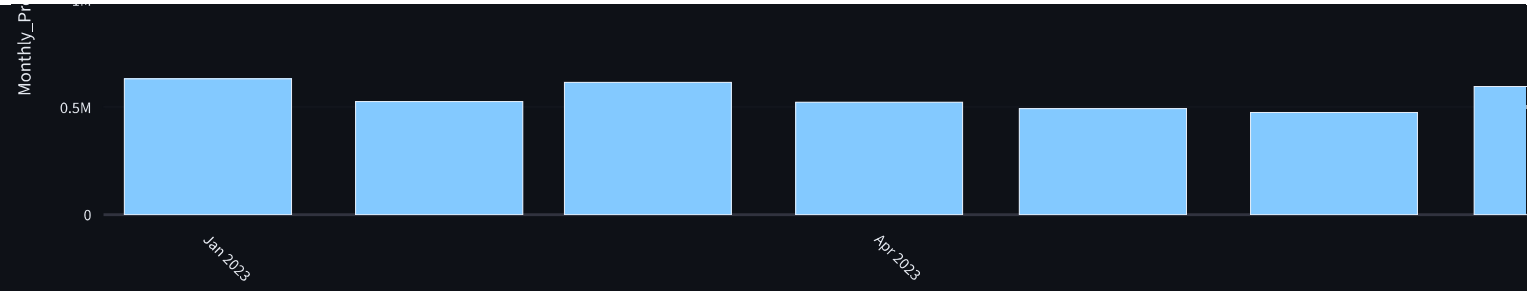
📈 Profit Trends Over Time

Daily Profit Trends



Monthly Profit Summary





💡 Key Profitability Insights

Total Profit	Avg Profit/Unit	Total Units Sold	Overall Margin
$22,025,469.03	$27.41	791,960	40.0%

🥇 **Most Profitable SKU:** TEX001 ($306,092.69)

🧵 **Most Profitable Material:** tencel

🏷️ **Most Profitable Category:** jacket

🏬 **Top Performing Store:** NYC_Flagship

✅ Profitability analysis completed!

⚠️ Prediction failed: The feature names should match those that were passed during fit. Feature names unseen at fit time:

- customer_rating
- initial_stock
- on_promotion
- remaining_stock
- store_size
- ...

📊 Data analysis charts are still available above

📥 Export Results

📊 Download Analysis CSV

💰 Download Profitability Analysis

📋 Download Summary